

Welcome to the 911 Emergency Calls Data Analysis Project. To begin, you will need to download the "911.csv" dataset from Kaggle (search for the MontcoAlert 911 dataset). Or Just simply paste the below code in your IDE. Once your environment is set up, answer the following questions sequentially.

```
import kagglehub

import pandas as pd

import numpy as np

import os

import matplotlib.pyplot as plt

import seaborn as sns

# Tell pandas to show all columns and expand the width limit
'''
By default, pandas will automatically compress or truncate your data tables
if they exceed the standard width of your IDE console, hiding valuable
columns behind ellipses (...). To prevent this and force Python to display
your entire dataset cleanly, include these two lines at the very top of your
script:
'''

pd.set_option('display.max_columns', None)

pd.set_option('display.width', 1000)

# Download dataset to local cache

path = kagglehub.dataset_download("mchirico/montcoalert")

# Point pandas to the CSV inside the cached folder

csv_file = os.path.join(path, "911.csv")

# Load the data into a DataFrame

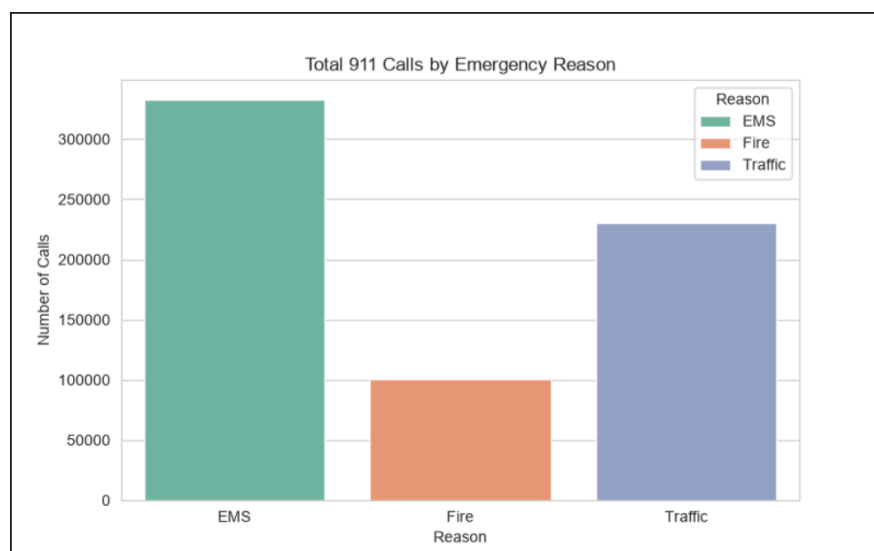
df = pd.read_csv(csv_file)
```

Basic Data Exploration

- **Question 1:** What are the top 5 zip codes for 911 calls?
- **Question 2:** What are the top 5 townships (`twp`) for 911 calls?
- **Question 3:** How many unique title codes exist in the dataset?
- **Question 4:** What is the full array/list of all unique title codes?

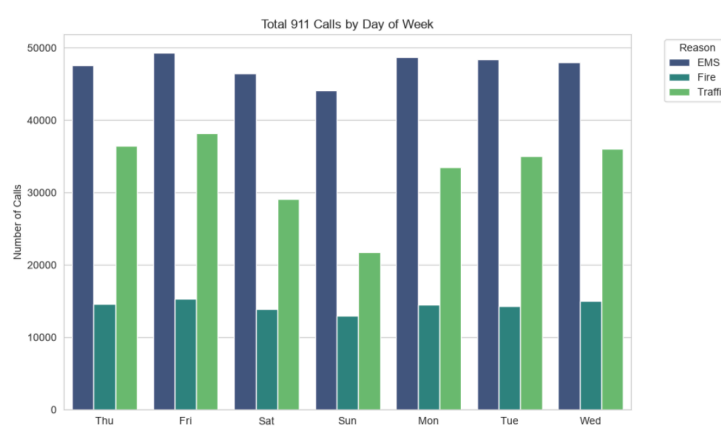
Feature Engineering & Categorical Visualization

- **Question 5A:** In the `title` column, there are "Reasons/Departments" specified before the colon (e.g., "EMS: BACK PAINS/INJURY"). Create a new column called `Reason` that contains only this string value. Based on this new column, what are the three most common reasons for a 911 call?
- **Question 5B:** Use Seaborn to create a countplot of 911 calls separated by this new `Reason` column. Your output should match the graph shown below:

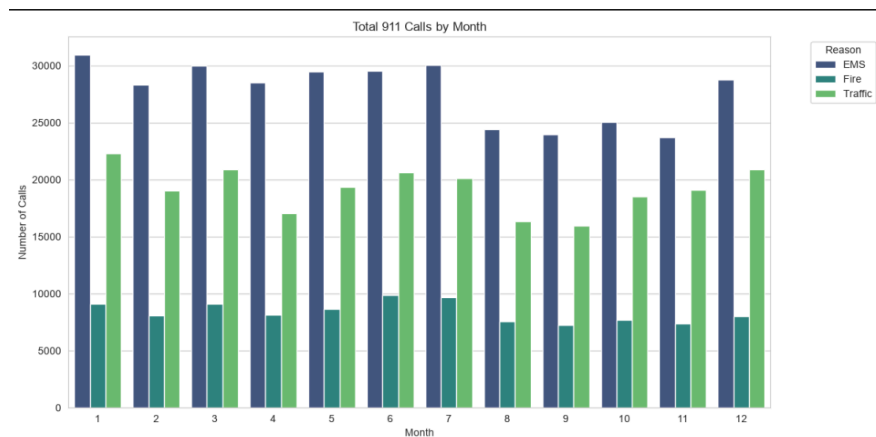


Time-Series Parsing & Visualization

- **Data Preparation:** What is the data type of the objects in the `timeStamp` column? Convert this column from strings to actual Datetime objects. Then, create three new columns called `Hour`, `Month`, and `Day of Week` based on the `timeStamp` column. Map the `Day of Week` integers to actual string names (e.g., 'Mon', 'Tue').
- **Question 6A:** Use Seaborn to create a countplot of the `Day of Week` column, coloring the bars based on the `Reason` column. Recreate the graph shown below:

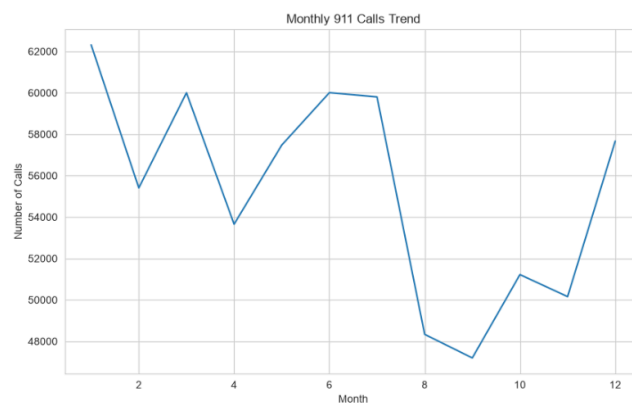


- **Question 6B:** Do the exact same operations for the **Month** column to recreate the following graph:

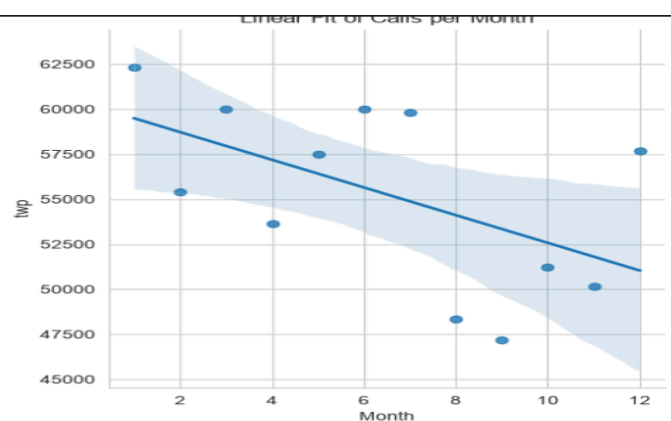


Aggregation, Regression & Daily Trends

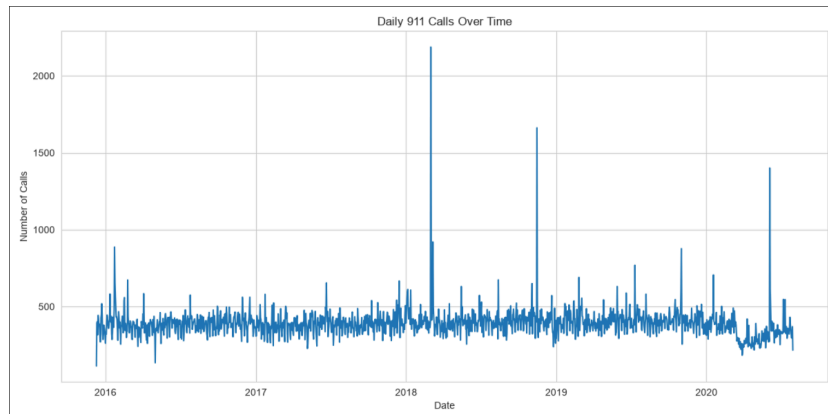
- **Question 7A:** Group the data by the **Month** column, aggregate it using the count method, and create a simple line plot indicating the count of calls per month to match the visualization below:



- **Question 7B:** Use Seaborn's **lmplot()** to create a linear fit on the number of calls per month. Recreate the graph shown below:

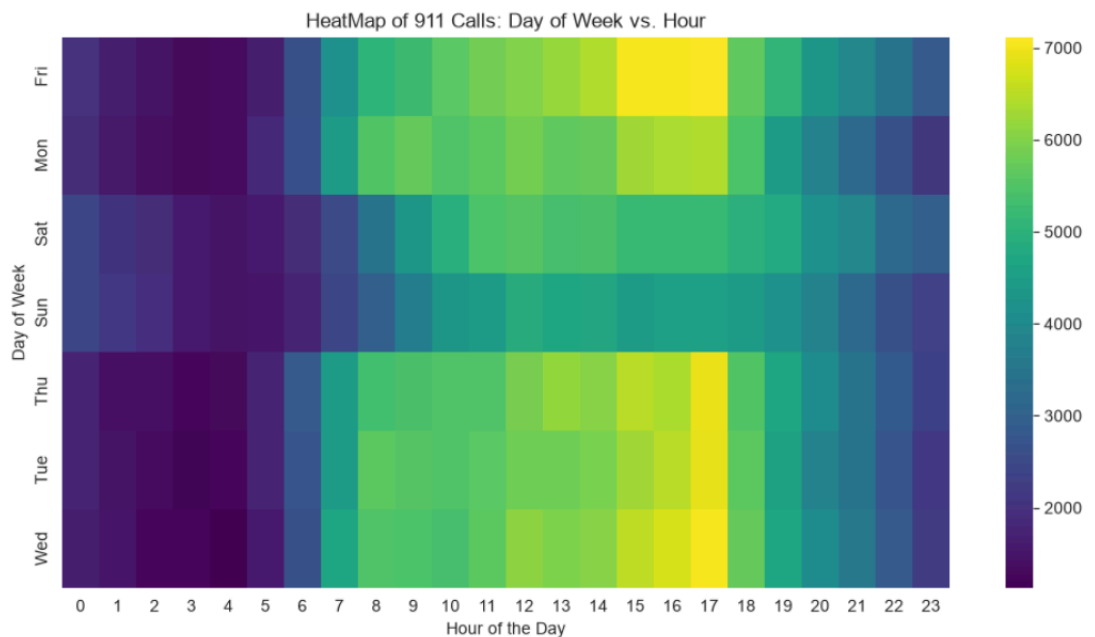


- **Question 7C:** Create a new column called `Date` that contains the date extracted from the `timeStamp` column. Group the data by this `Date` column and create a time-series line plot showing the counts of 911 calls over time, matching the graph below:



Pivot Tables & Heatmaps

- **Question 8:** Restructure the dataframe so that the columns become the Hours of the day and the index becomes the Day of the Week. Use this new DataFrame to create a Heatmap with Seaborn that matches the reference image below:



Hierarchical Clustering

- **Question 9:** Using the exact same Day of Week vs. Hour matrix created in Question 8, create a Clustermap to group similar days and hours together. Recreate the clustermap shown below: *(Insert Graph 9 Image Here)*

